

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Beat Optical, and Does It Hold Across ViT and CNN?

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Abstract

Detecting “dark” vessels—ships not broadcasting their Automatic Identification System (AIS) signal—in spaceborne synthetic aperture radar (SAR) is operationally important and label-scarce. Pretrained backbones can help, but it is unclear which pretraining recipe transfers best, and whether the answer is the same for Transformer and convolutional backbones. We propose a controlled label-efficiency study on xView3-SAR [8] that holds one point-native detector fixed and varies only the downloaded backbone initialization, across two size-matched tracks (ViT-B/16, ~86M; ConvNeXt-V2-Base, ~89M) and four pretraining recipes each: random, optical, SAR, and supervised out-of-domain SAR. Because every backbone is a released checkpoint, differences are attributable to the pretraining source rather than to our own optimization choices.

1. Introduction

Detecting ships not broadcasting their AIS signature—“dark vessels”—in spaceborne SAR is a critical maritime-monitoring problem with scarce labels: xView3-SAR [8] provides Sentinel-1 imagery, but its human-verified dark-vessel labels are few, so supervised-only training overfits.

Pretrained backbones can help, but it is unclear which pretraining recipe transfers best: self-supervision on generic optical satellite imagery (fMoW-RGB [2] via SatDINO [9]), self-supervision on large-scale SAR (SAR-1M via SARMAE [7]), or supervised detection on labeled out-of-domain SAR (LS-SSDD [11])—and whether the best recipe is the same for a ViT [4] and a ConvNeXt [10]. We compare released checkpoints rather than pretraining our own so that each arm faithfully represents its recipe as pub-

lished; re-pretraining would confound the pretraining source with our own optimization choices [13].

With the detector, schedule, and input fixed, we ask: (Q1) how much labeled xView3 data does each recipe save relative to random initialization to reach a target dark-vessel F1; (Q2) does SAR-domain pretraining beat optical in the scarce-label regime; and (Q3) is that ordering architecture-general, holding for both a ViT and a size-matched CNN? We hypothesize that, within each track, SAR-domain pretraining yields the steepest low-label gains (random < optical $\lesssim$ supervised < SAR), and that this ordering holds for both architectures.

2. Approach

Design. The study has two architecture-matched tracks—ViT-B/16 (~86M) and ConvNeXt-V2-Base (~89M, deliberately size-matched)—each with four arms that share the same head, optimizer, schedule, and seeds; only the downloaded initialization differs (Table 1). Initialization is the only variable within a track and architecture the only variable across matched roles, so comparing arm k to arm $k+4$ isolates the architecture family with the pretraining role held fixed.

Pretraining regimes. The recipes differ in modality, labels, and domain: optical (SatDINO, DINO [1]; BigEarthNet-S2) is a different sensing modality (no speckle, different scattering physics); SAR, self-supervised (SARMAE, an MAE [5]; BigEarthNet-S1) is the same modality but pretrained without detection labels on a different scene distribution; and labeled out-of-domain SAR (LS-SSDD) is the same modality and a related supervised task, but a different acquisition. Q2 thus contrasts “same pixels, different task” against “different pixels, same task.”

Arm	Track	Backbone	Initialization	Params	On curve?	Role
1	ViT	ViT-B/16	random init	~86M	yes	ViT floor
2	ViT	ViT-B/16	SatDINO (fMoW-RGB, DINO)	~86M	yes	optical-domain transfer
3	ViT	ViT-B/16	SARMAE (SAR-1M, MAE)	~86M	yes	SAR-domain transfer
4	ViT	ViT-B/16	LS-SSDD supervised (we train)	~86M	yes	labeled out-of-domain SAR
5	CNN	ConvNeXt-V2-B	random init	~89M	yes	CNN floor
6	CNN	ConvNeXt-V2-B	BigEarthNet-S2 (optical RS)	~89M	yes	optical-domain transfer
7	CNN	ConvNeXt-V2-B	BigEarthNet-S1 (SAR RS)	~89M	yes	SAR-domain transfer
8	CNN	ConvNeXt-V2-B	LS-SSDD supervised (we train)	~89M	yes	labeled out-of-domain SAR
R1	ref	ConvNeXt-V2-B	ImageNet (contingent)	~89M	no	generic-transfer reference
R2	ref	YOLO26	COCO-pretrained	external	no	detector reference
R3	ref	LocateAnything-3B	zero-shot	~3B	no	VLM reference

Table 1. Eight controlled arms in two size-matched tracks (~86M vs. ~89M), plus three external references reported off the curves. Optical anchors are SatDINO [9] (ViT) and BigEarthNet-S2 [3] (CNN); SAR anchors are SARMAE [7] (ViT) and BigEarthNet-S1 [3] (CNN).

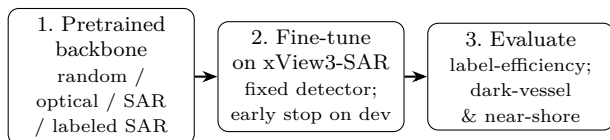


Figure 1. The three stages, run identically for every arm; only stage 1 (the downloaded backbone) varies within a track.

Detector. xView3 labels are points and its metric matches predictions by distance, not box IoU, so we use a point-native heatmap head in the style of CenterNet / Objects as Points [12]: each vessel is a Gaussian blob on a stride-4 map, decoded by peak-finding with distance-based non-maximum suppression. Every arm receives the same fixed 3-channel input [VH, VV, VH-VV] in dB, so input handling is never a confound between arms.

Fine-tuning and comparability. All arms fine-tune end-to-end through one code path with shared AdamW, augmentation, and seeds, and are trained until dev-split F1 stops improving (early stopping), not for a fixed epoch budget; the decision threshold is selected on dev and frozen before test. Continuous-integration guard tests make comparability a checked invariant rather than a claim: a backbone-parity test asserts identical within-track parameter counts and matching head-adaptor geometry, a checkpoint-load test asserts each backbone loads its pretrained weights (not silently random), and a scorer-immutability test freezes the evaluation code (Figure 1).

3. Data and Evaluation

We use xView3-SAR [8] via the SARFish mirror (Sentinel-1 GRD, dual-polarization VH/VV): 75–150 training scenes chipped at 800×800 ($\sim 10^4$ chips), sub-sampled at the scene level to 10/25/50/100% of labels and split 75/15/10 (train/dev/test); the ~50 human-verified scenes are a disjoint, once-touched final-evaluation set. Arms 4 and 8 pretrain on LS-SSDD-v1.0 [11] (~9,000 labeled 800×800 sub-images), used only as a pretraining source and never mixed into the xView3 splits. The headline result is a two-track label-efficiency curve (detection F1 vs. 10/25/50/100% labels; eight lines) plus dark-vessel recall and near-shore F1, the operationally hardest slices: the low-label gap over the random floor answers Q1–Q2, and matched-role gaps across tracks answer Q3. External references (YOLO26 [6], a zero-shot VLM, and a contingent ImageNet-ConvNeXt) situate the eight arms without entering the controlled comparison.

4. Timeline and Feasibility

Two contributors work independent lanes (data/scorer; detector/eval) with no pretraining handoff to block on. Because all foundation-model backbones are downloaded, compute is modest: the full eight-arm grid with seed reruns plus the two small LS-SSDD supervised backbones is ~293–360 GPU-hours (~3–4 nights on one $8 \times V100$ node)—we download checkpoints for comparability of recipes as published, and the low cost is a side benefit. The downloaded-backbone arms run first to de-risk the fixed-input channel format. We target the WACV 2027 Round-2 deadline (paper registration August 21, 2026; submission August 28, 2026, AoE).

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